

SPACE ARCHAEOLOGY

VR Experience

Space Archaeology VR lets people play a role from future and travel through the universe as a space explorer. In this virtual experience, user need to search for the remains of human life — tiny fragments, objects, and signals left in history. Each discovery represent a small piece of history. By combining narrative exploration with interactive gameplay, this project lets user enjoy the large-scale and ambitious by providing micro story telling.

INSPIRATION



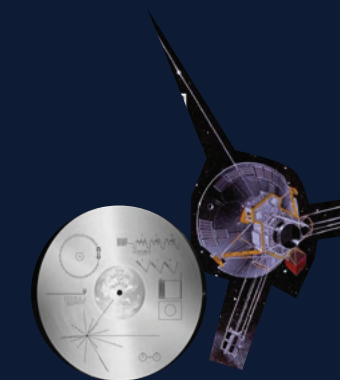
I've been seeing a growing number of news stories about space travel recently, and many of my friends and I are deeply fascinated by it. However, as this technology is still in its early stages, only a small number of people can actually experience it firsthand. At the same time, countless science fiction writers and scientists have turned their attention and creativity toward the cosmos, and I find their work incredibly inspiring. That's why I want to use a technology that's already widely accessible—VR—to bring the beauty and wonder of the universe closer to everyone.

RESEARCH

success



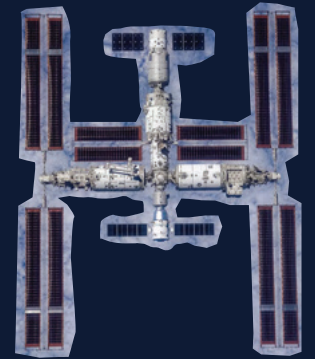
1942 V-2 Rocket
world's first guided ballistic missile



1977 – Voyager 1
the farthest human-made object



2001 – Dennis Tito,
First Space Tourist



2021 – Chinese Space Station
long-term human spaceflight

1957 – Laika
the Space Dog

1986 – Challenger Disaster
exploded 73 seconds after liftoff

1991 – Sergei Krikalev's
Record Mission

unexpected



National Prestige and Scientific Pioneering



2022 – first fully private crew to the orbiting complex



Before the 2000s, space exploration was largely driven by government agencies and framed by geopolitical competition, especially during the Cold War. Milestones such as the launch of Sputnik, the first human in space, and the Apollo Moon landings symbolized national pride and technological supremacy. Scientific discovery and military strategy were the main motivations, and ordinary people could only observe from afar. However, as the 21st century began, a new chapter unfolded

The Rise of Commercial and Public Spaceflight



2025 – Blue Origin's All-female
Flight

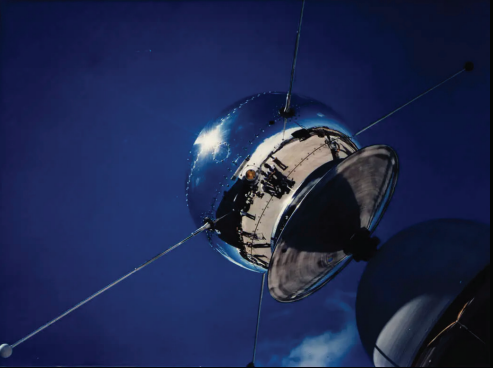


Blue Origin launch of tourism
rocket ends nearly 2-year
hiatus

Since the early 2000s, space travel has undergone a remarkable transformation—from a government-led endeavor limited to professional astronauts to an increasingly commercialized, democratized, and international pursuit. What was once science fiction has gradually become reality, as private companies like SpaceX, Blue Origin, and Virgin Galactic have pioneered technologies that open space to civilians. This shift marks a new era in which space is no longer the exclusive domain of national agencies, but a frontier where tourism, research, and global collaboration coexist, shaping a future in which access to space is more inclusive and driven by innovation across sectors.

INFLUENCE

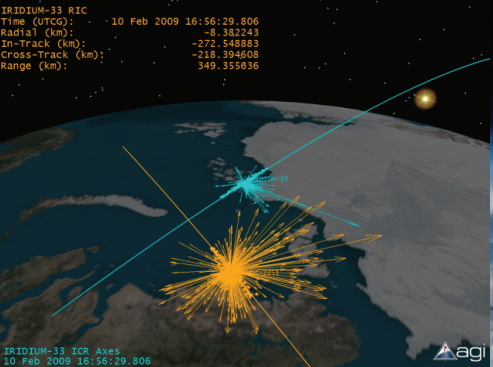
Vanguard 1
One of the oldest space debris



Jerry Linenger glove
fell during the maintenance of the
space station



Iridium 33 vs Cosmos 2251
About 2,300 traceable fragments
were produced



Outer Space Treaty
The most fundamental legal
document of space law



Defunct Satellites

Source: Abandoned or failed spacecraft.
Impact: Crowded orbits raise collision
risks and endanger active missions.

Lost Tools in Space

Source: Dropped by astronauts during
extravehicular repairs.
Impact: Small but fast debris that
threatens spacecraft and ISS.

Collision/Explosion

Source: Fuel residue explosions or
satellite collisions.
Impact: Rapid debris multiplication;
long-term threat to space operations.

Space Ethics & Resource

Source: Moon, Mars, and asteroid
resource exploitation.
Impact: Legal uncertainty, risk of
conflict, and environmental harm.

USER RESEARCH

Attitude:

What is your opinion and perspective on space
travel?

Impact:

What impacts do you think space travel could have?

Experience:

If you could go on a space trip, what would you
expect to experience?

Concerns:

What worries do you have about space travel?



Bill expresses enthusiastic admiration, seeing space travel as a lifelong dream. In contrast, Joy reveals a conflicted perspective, worried about repeating environmental mistakes in space. These views reflect a tension between aspiration and ethical concern.

Bill: "I've been fascinated by the universe since I was a kid... It's one of my biggest dreams."
Joy: "To be honest, I feel pretty conflicted about space travel... I don't want us to do the same harm to the universe."

Robert focuses on negative environmental impacts, warning about space debris. Ella highlights cultural and symbolic impacts, seeing even lost objects as pieces of collective human memory. These responses show that impact can be both practical and emotional.

Robert: "If we keep sending stuff into space, I worry we might create new kinds of space pollution..."
Ella: "It's kind of amazing to think how a single object—like a glove—can become part of space history."

Bill is excited by the physical and visual thrill of space travel. Joy anticipates a deeply emotional and philosophical experience. Their expectations show how space can evoke both awe and inner transformation.

Bill: "I would love to experience the feeling of weightlessness, seeing Earth from space—that would be life-changing."
Joy: "Honestly, I'd probably be really emotional. Just imagining the vastness of space makes me feel small but connected."

Robert's concern is about personal safety and the risks of space travel. Ella worries about repeating Earth's mistakes, transferring human pollution and behavior into space. These concerns highlight fears around risk, responsibility, and repetition of history.

Robert: "I don't think I'd go unless it felt truly safe... Space seems pretty risky to me."
Ella: "We've made a mess on Earth, and now we're sending things into space. That worries me—we might just be exporting our bad habits."

curiosity

safety concerns

ethical problem

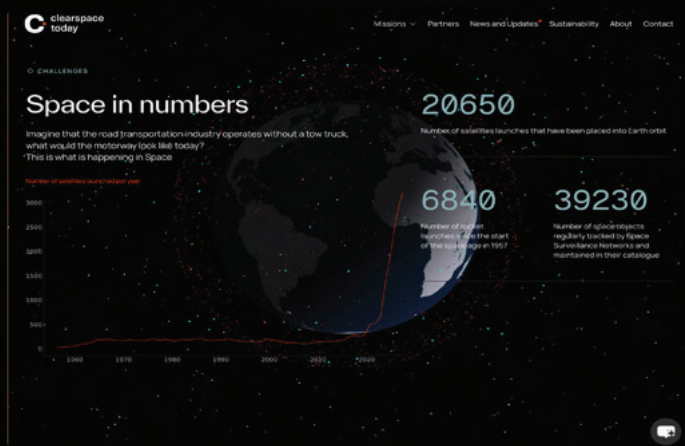
space debris

CASE STUDY



AIRPANO

This virtual reality project offers users a first-person immersive tour of Earth’s most stunning landscapes and fragile ecosystems.



Clearspace Today

This is a project designed to collect space debris and repurpose it into functional components, addressing the growing issue of space junk.

Inspired by the two projects, I want to create a VR space travel experience where users can journey through the future of outer space and encounter the legacy left behind by past human activities—such as space debris from specific events or abandoned satellites. The experience will gradually zoom out, helping viewers realize that just as we have impacted Earth, our presence in the universe also leaves a footprint. Through this, I hope to spark reflection: if we dream of future space travel, how can we ensure it doesn’t repeat the mistakes we made on Earth? What responsibilities come with expanding into space?



MINDMAP

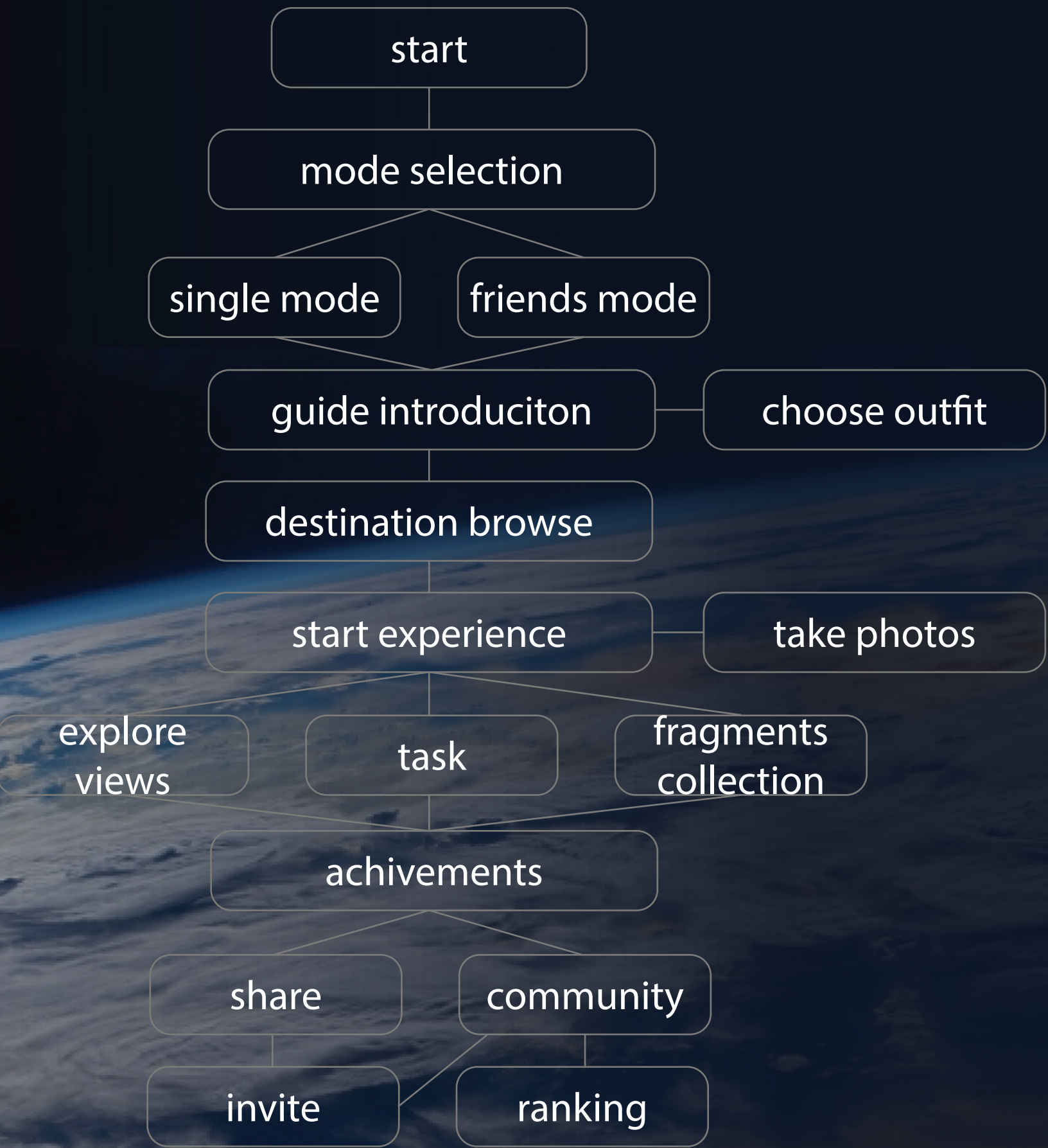


Humanity’s growing impact on space, such as debris and abandoned satellites, often goes unnoticed by the public. This VR project creates an immersive journey through space where users encounter the legacies of past human activities. The goal is to inspire reflection on our responsibility for the universe and encourage visions of a more sustainable future for space travel.

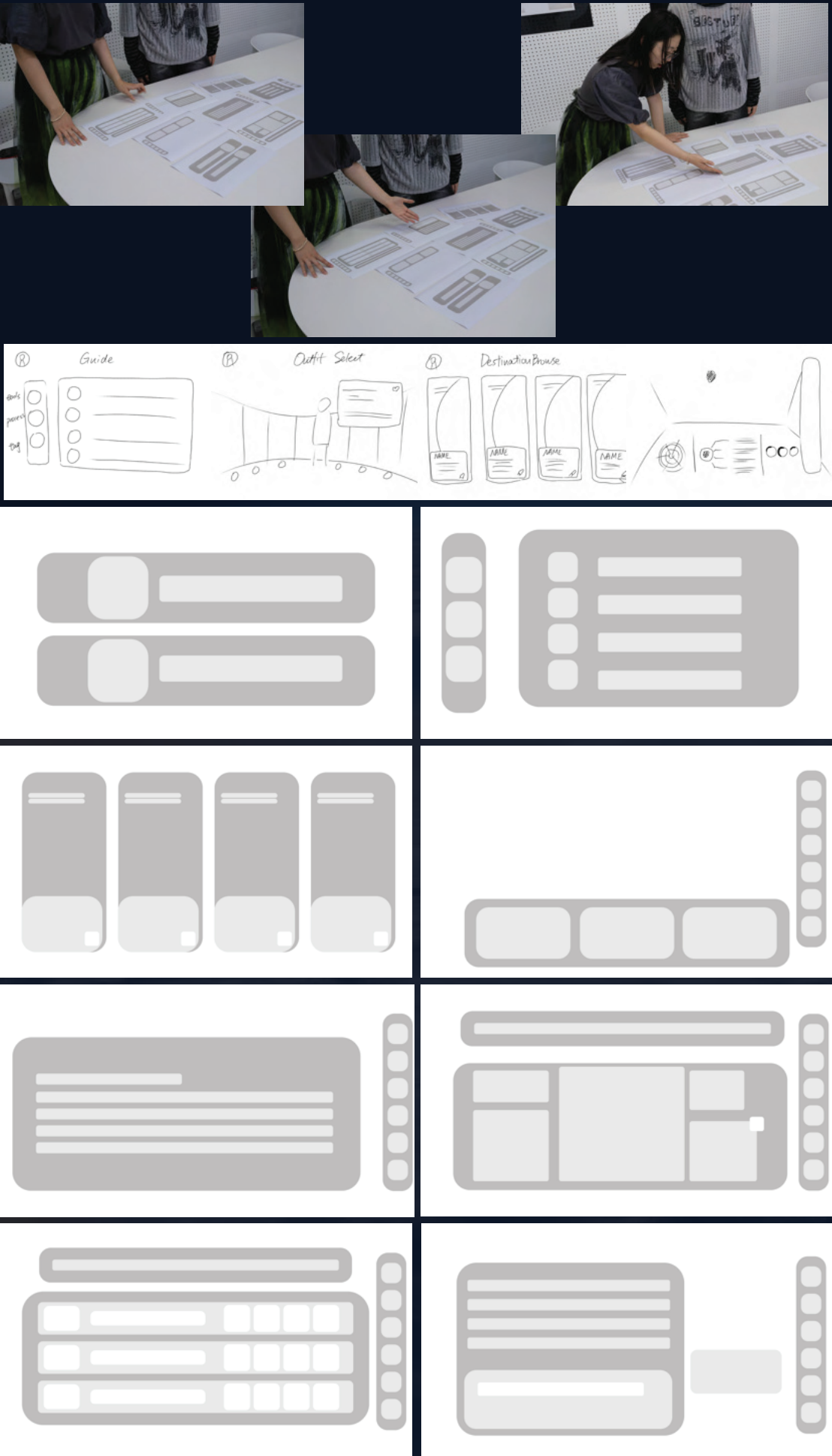
CONCEPT



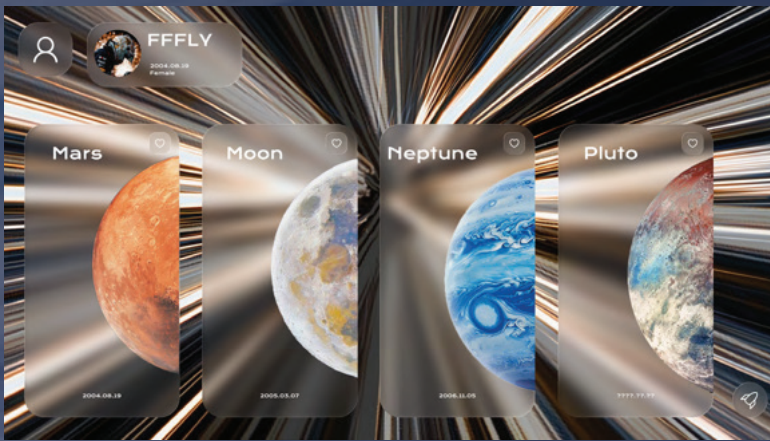
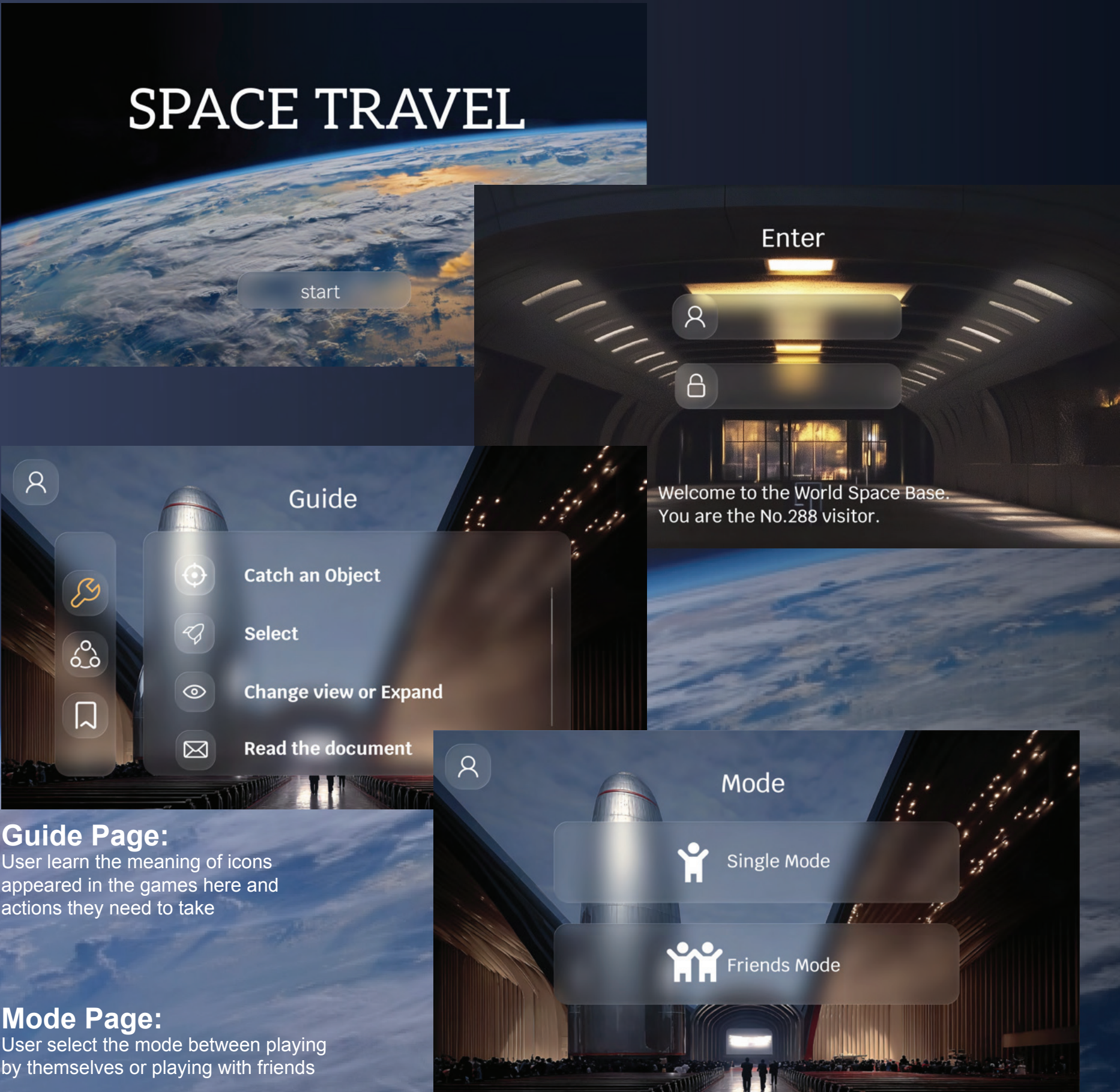
INFOMATION ARTICHECTURE



WIREFRAM



VR INTERFACE DESIGN



Travel Page:

Users select their outfits and then choose their destination. They can also save certain destinations for future travels. Each destination offers a different visual experience. During the VR session, users can explore a full 360° scene. They can also receive tasks and read descriptions on the control console.



Task- Catch Fragment:

Users can pick up fragments and read the related information and stories on the control console. The control console features a progress bar that tracks the collected fragments and encourages users to keep playing.

